

Cousin and Sexy Prime Pairs on the $6N$ Skeleton: Three ω -Distortions Set by Pairing Geometry

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Abstract

Parts I–VIII built a conditional theory of twin pairs $(6N-1, 6N+1)$ on the $6N$ skeleton, in which the pair rate rises sharply with the factor count $\omega_{>3}(N)$ of the single shared centre. Here we ask whether the same ω -dependence carries over to cousin pairs (difference 4) and sexy pairs (difference 6). It does not carry over identically; instead it distorts into three distinct, geometry-determined shapes. On the $6N$ skeleton a cousin pair is $(6N+1, 6(N+1)-1)$ and a sexy pair is either $(6N-1, 6(N+1)-1)$ (“sexy-A”) or $(6N+1, 6(N+1)+1)$ (“sexy-B”); unlike the twin, all of these straddle two consecutive centres N and $N+1$. Binning by the left centre’s ω on the 1.5×10^9 centres of S_{10} (over 5×10^8 pairs), we find: the twin rate rises monotonically (to $4.8\times$ at $\omega = 7$); the cousin and sexy-A rates fall monotonically (to $0.13\text{--}0.15\times$) and are numerically identical to three decimals at every ω ; and the sexy-B rate is non-monotone, peaking near $\omega = 4$ before falling. The cousin/sexy-A coincidence is explained by their shared right member $6(N+1)-1$. The single-centre enrichment of the twin thus reverses or becomes non-monotone once the pair straddles two centres, with the sign of the distortion fixed by which wings are involved. We give the phenomenon and its geometric account; a quantitative two-centre mechanism is posed as the open problem. No claim is made about the infinitude of any prime constellation.

1 Geometry of the pairs

On the $6N$ skeleton (all primes > 3 lie on $6N \pm 1$) the three constellations take fixed forms. A twin pair is $(6N-1, 6N+1)$, the two wings of a *single* centre N . A cousin pair (difference 4) must be $(6N+1, 6N+5) = (6N+1, 6(N+1)-1)$, since the alternative $6N+3$ is always divisible by 3; it straddles centres N and $N+1$ (right wing of N , left wing of $N+1$). A sexy pair (difference 6) is either

$$\text{sexy-A: } (6N-1, 6(N+1)-1) \quad \text{or} \quad \text{sexy-B: } (6N+1, 6(N+1)+1),$$

both straddling N and $N+1$ (two left wings, or two right wings, respectively). The essential contrast: the twin lives on one centre, while cousin and sexy live on two consecutive centres. The conditional theory of Parts I–VIII attributes a pair to a centre’s factor count $\omega_{>3}$; for the straddling pairs we attribute to the *left* centre N , the natural analogue.

2 Three distortions in S_{10}

We scan all 1.5×10^9 centres of S_{10} ($6N \in [10^9, 10^{10})$), determine $\omega_{>3}(N)$ and the primality of both wings of every centre, and count each constellation binned by the left centre’s ω (Table 1, Fig. 1). Rates are normalised to $\omega = 1$ to compare shapes.

ω	left centres	twin	cousin	sexy-A	sexy-B
1	235,410,094	1.000	1.000	1.000	1.000
2	564,607,456	1.196	0.962	0.962	1.077
3	484,576,391	1.495	0.901	0.901	1.157
4	183,629,904	1.936	0.804	0.804	1.198
5	30,002,229	2.581	0.648	0.650	1.134
6	1,751,627	3.524	0.420	0.414	0.881
7	22,229	4.808	0.127	0.150	0.402

Table 1: Pair rate by left-centre $\omega_{>3}$, normalised to $\omega = 1$, on S_{10} . Twin rises monotonically; cousin and sexy-A fall monotonically and coincide to three decimals; sexy-B peaks near $\omega = 4$ then falls.

Cousin and sexy prime pairs in S_{10} (15B centres): ω -dependence distorts by pairing geometry

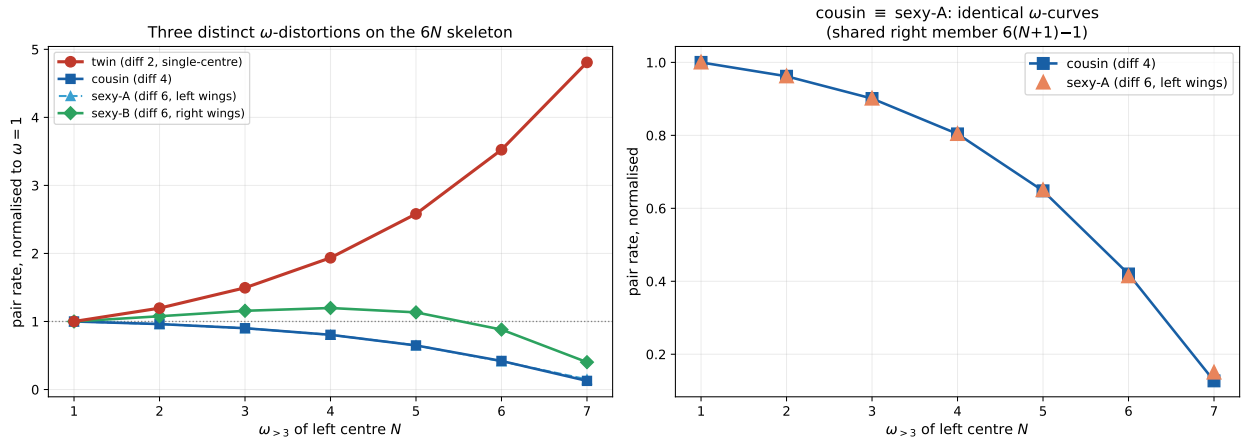


Figure 1: Left: the three distortions. Twin (red) rises; cousin and sexy-A (blue, light) fall and overlap; sexy-B (green) is non-monotone. Right: cousin and sexy-A curves coincide at every ω , explained by their shared right member $6(N+1)-1$.

Twin: monotone rise. The twin reproduces the Parts II–VIII enrichment: both wings sit on the single centre N , so a factor-rich N protects both, and the rate climbs to $4.8\times$ at $\omega = 7$.

Cousin and sexy-A: monotone fall, and identical. Both fall to $0.13\text{--}0.15\times$ at $\omega = 7$ —opposite to the twin—and they are numerically equal at every ω (e.g. 0.804 vs 0.804 at $\omega = 4$, 0.420 vs 0.414 at $\omega = 6$). The coincidence has a clean cause: cousin is $(6N+1, 6(N+1)-1)$ and sexy-A is $(6N-1, 6(N+1)-1)$, sharing the right member $6(N+1)-1$. After normalisation the left member’s own ω -response divides out, leaving the response of the shared right member on the *adjacent* centre $N+1$; that is common to both, so the curves coincide. The fall (rather than rise) reflects that the right member sits on $N+1$, not N : a factor-rich left centre N does not protect a prime on the neighbouring centre, and the conditioning instead correlates with suppression.

Sexy-B: non-monotone. Sexy-B rises to a peak near $\omega = 4$ ($1.20\times$) and then falls ($0.40\times$ at $\omega = 7$), a third shape distinct from both others. It is the two-right-wing constellation $(6N+1, 6(N+1)+1)$, and its non-monotonicity suggests two competing effects—an early enrichment-like gain and a later straddle suppression—that we do not separate here.

3 Discussion and open problem

The single-centre twin enrichment does not generalise unchanged: once a pair straddles two consecutive centres, the ω -dependence of the left centre distorts in a sign and shape fixed by which wings the pair occupies—falling for cousin and sexy-A (whose conditioned member’s partner lives on $N+1$), non-monotone for sexy-B. This is the expected qualitative consequence of the two-centre geometry that Parts V–VIII analysed for the twin’s own neighbour correlations, now appearing at leading order because the constellation itself is two-centred.

Open problem. Predict the three curves quantitatively from a two-centre conditional model over $(N, N+1)$, using the modular-shift/lockdown mechanism of Part V: the rise of twin, the monotone fall and exact coincidence of cousin and sexy-A, and the non-monotone peak of sexy-B. A successful model would express each constellation’s ω -curve through the admissibility of its two members across the two centres.

4 Conclusion

Cousin and sexy prime pairs on the $6N$ skeleton do not inherit the twin’s monotone ω -enrichment. Measured on S_{10} , they split into three geometry-determined shapes: the single-centre twin rises, the straddling cousin and sexy-A fall and coincide (shared right member), and sexy-B is non-monotone. The distortion is governed entirely by the pairing geometry—which centres and which wings—rather than by the difference alone. We report the phenomenon and its geometric account and leave the quantitative two-centre mechanism open. As throughout this series, no claim is made about the infinitude of twin, cousin, or sexy primes, or any k -tuple conjecture; this is a measured, geometry-resolved account of conditional pair rates on the $6N$ skeleton.

Data and code availability. The scan and the per- ω counts are openly available [6]; the S_{10} twin count 23,988,173 matches Part I.

References

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